Assignment-02

1)

	Time t (sec)	Velocity (m/s)
x_0	2	10
x_1	4	20
x_2	6	25

nodes = 3
degree = 2

a) $P_2(x_0) = a_0 + a_1 x_0 + a_2 x_0^2 = f(x_0)$
$P_2(x_1) = a_0 + a_1 x_1 + a_2 x_1^2 = f(x_1)$
$P_2(x_2) = a_0 + a_1 x_2 + a_2 x_2^2 = f(x_2)$

$$\begin{bmatrix} 1 & 2 & 2^2 \\ 1 & 4 & 4^2 \\ 1 & 6 & 6^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 6 & 36 \end{bmatrix}^{-1} \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 1 \\ -1.25 & 2 & -0.75 \\ 0.125 & -0.25 & 0.125 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -5 \\ 8.75 \\ -0.625 \end{bmatrix}$$

$\cancel{P_2(x) = -5x + 8}$

$P_2(x) = -5 + 8.75x - 0.625x^2$

We got the interpolating polynomial for velocity, now we need to differentiate to find acceleration.

$$P_2'(x) = 8.75 - (0.625 \times 2)(x)$$
$$P_2'(x) = 8.75 - 1.25x$$

At time = 7 secs,

$$P_2'(x) = 8.75 - (1.25 \times 7)$$
$$P_2'(x) = 0 \ m/s^2$$

Approximate value of acceleration is $0 m/s^2$.

(b) Using Lagrange Method,

$$P_2(x) = f(x_0) l_0(x) + f(x_1) l_1(x) + f(x_2) l_2(x).$$

$$l_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$l_0(x) = \frac{(x-4)(x-6)}{(2-4)(2-6)}$$

$$l_0(x) = \frac{(x-4)(x-6)}{8}$$

$$l_0(x) = \frac{1}{8}(x^2 - 6x - 4x + 24)$$

$$l_0(x) = \frac{1}{8}(x^2 - 10x + 24)$$

$$l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$l_1(x) = \frac{(x-2)(x-6)}{(4-2)(4-6)}$$

$$l_1(x) = -\frac{1}{4}(x^2 - 6x - 2x + 12)$$

$$l_1(x) = -\frac{1}{4}(x^2 - 8x + 12)$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

$$l_2(x) = \frac{(x-2)(x-4)}{(6-2)(6-4)}$$

$$l_2(x) = \frac{1}{8}(x^2-4x-2x+8)$$

$$l_2(x) = \frac{1}{8}(x^2-6x+8)$$

$$P_2(x) = \frac{10}{8}(x^2-10x+24) + (-5)(x^2-8x+12)$$

$$+\frac{25}{8}(x^2-6x+8).$$

ⓒ If a new data point is added in the above scenario, then it would be better to use Newton's Divided Difference Method. This would help to avoid recomputation of Lagrange form.

If a new node is added then there would be 4 nodes so the degree of the new polynomial would be 3.

②	x	$f(x)$
x_0	$-\pi/2$	$\pi/2$
x_1	0	0
x_2	$\pi/2$	$\pi/2$

nodes = 3, degree = 2

$f(x) = x\sin(x)$

a) $P_2(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1)$

$x_0 = -\pi/2 \quad f[x_0] = \pi/2$

$f[x_0, x_1] = \dfrac{f[x_1]-f[x_0]}{x_1 - x_0}$

$x_1 = 0 \quad f[x_1] = 0$ $\quad f[x_0, x_1, x_2]$

$f[x_1, x_2] = \dfrac{f[x_2]-f[x_1]}{x_2 - x_1}$

$x_2 = \pi/2 \quad f[x_2] = \pi/2$

$$f[x_0, x_1] = \frac{0-\pi/2}{0+\pi/2} = -1$$

$$f[x_1, x_2] = \frac{\pi/2 - 0}{\pi/2 - 0} = 1.$$

$$f[x_0, x_1, x_2] = \frac{1-(-1)}{\pi/2-(-\pi/2)} = \frac{2}{\pi}.$$

$$P_2(x) = \pi/2 + (-1)(x+\pi/2) + \left(\frac{2}{\pi}\right)(x+\pi/2)(x)$$

(b) At $x = \pi/4$,

$$P_2(\pi/4) = \pi/2 + (-1)(\pi/4 + \pi/2) + \left(\frac{2}{\pi}\right)(\pi/4 + \pi/2)(\pi/4)$$

$$P_2(\pi/4) = \frac{\pi}{8}.$$

(c) If a new node π is added to the above nodes, then the degree of the interpolating polynomial would be 3.

$$P_3(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$
$$+ f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2)$$

$x_0 = -\frac{\pi}{2}$	$f[x_0] = \frac{\pi}{2}$			
		$f[x_0, x_1]$		
$x_1 = 0$	$f[x_1] = 0$		$f[x_0, x_1, x_2]$	
		$f[x_1, x_2]$		$f[x_0, x_1, x_2, x_3]$
$x_2 = \frac{\pi}{2}$	$f[x_2] = \frac{\pi}{2}$		$f[x_1, x_2, x_3]$	
		$f[x_2, x_3]$		
$x_3 = \pi$	$f[x_3] = 0$			

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2}$$
$$= \left(0 - \frac{\pi}{2}\right) / (\pi - \pi/2)$$
$$= -1$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$$
$$= \frac{-1 - 1}{\pi - 0} = -\frac{2}{\pi}.$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$$

$$= \frac{\left(-\frac{2}{\pi} - \frac{2}{\pi}\right)}{\left(\pi - \left(-\frac{\pi}{2}\right)\right)}$$

$$= -0.27019$$

$$P_3(x) = \frac{\pi}{2} + (-1)\left(x + \frac{\pi}{2}\right) + \left(\frac{2}{\pi}\right)\left(x + \frac{\pi}{2}\right)(x) + (-0.27019)\left(x + \frac{\pi}{2}\right)(x)\left(x - \frac{\pi}{2}\right)$$

3.

	x	$f(x)$
x_0	1	0
x_1	3	3.2958

$$f(x) = x \ln(x)$$

$$P_1(x) = l_0(x) f(x_0) + l_1(x) f(x_1)$$

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{(x-3)}{(1-3)} = -\frac{1}{2}(x-3)$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{(x-1)}{(3-1)} = \frac{1}{2}(x-1)$$

$$P_1(x) = -\frac{1}{2}(x-3)(0) + \frac{1}{2}(x-1)(3.296)$$

4 significant figures

$$P_1(x) = +\frac{1}{2}(x-1)(3.296).$$

4. $f(x) = e^{3x} - e^{-3x}$

3 nodes $\rightarrow [-1, 0, 1]$
so degree will be 2.

$$|f(x) - P_2(x)| = \left| \frac{f^{(3)}(\xi)}{3!} \underbrace{(x+1)(x-0)(x-1)}_{w(x)} \right|$$

$$f(x) = e^{3x} - e^{-3x}$$
$$f'(x) = 3e^{3x} - (-3)e^{-3x}$$
$$f'(x) = 3e^{3x} + 3e^{-3x}$$
$$f^2(x) = 9e^{3x} + 3(-3)e^{-3x}$$
$$f^2(x) = 9e^{3x} - 9e^{-3x}$$
$$f^3(x) = 27e^{3x} - 9(-3)e^{-3x}$$
$$f^3(x) = 27e^{3x} + 27e^{-3x}.$$

$$|f(x) - P_2(x)| = \left| \frac{27e^{3(\xi)} + 27e^{-3(\xi)}}{3!} \right| \left| (x+1)(x)(x-1) \right|$$

$$w(x) = (x+1)(x)(x-1)$$
$$w(x) = (x^2+x)(x-1)$$
$$w(x) = x^3 - x^2 + x^2 - x$$
$$w(x) = x^3 - x$$
$$w'(x) = 3x^2 - 1.$$
$$w'(x) = 0$$
$$3x^2 - 1 = 0$$
$$3x^2 = 1$$
$$x^2 = \frac{1}{3}$$
$$x = \pm \frac{1}{\sqrt{3}}$$

x	$w(x)$
$\sqrt{1/3}$	$-\frac{2\sqrt{3}}{9} = -0.3849$
$-\sqrt{1/3}$	$\frac{2\sqrt{3}}{9} = 0.3849$
-1.5	$-\frac{15}{8} = -1.875$
1.5	$\frac{15}{8} = 1.875 \leftarrow$ Max value

For $\frac{f^3(\xi)}{3!}$ part we will use the interval $[-1.5, 1.5]$ to find the maximum value.

For $\xi = 1.5$,

$$\left| \frac{27e^{3(1.5)} + 27e^{-3(+1.5)}}{3!} \right|$$

$= 405.12708$

The same value is for $\xi = -1.5$ as well so it is the maximum value.

$$|f(x) - P_2(x)| = 405.12708 \times 1.875$$
$$= 759.613$$
$$= 759.6 \text{ [4 significant figures]}$$